

A Novel P4 Target Architecture for Runtime-Reconfigurable NDN Data Planes

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Motivation



- Switch functionalities are fixed and can only be changed by the switch manufacturer.
- P4 opens up the data plane to allow network operators to specify their custom forwarding behaviors using P4 programs.
- Current P4 architectures allow a single large P4 program to run on a switch. Hence, downtime is needed for updates.
- P4 has limitations when processing string-based protocols as in the case of NDN.



Objectives



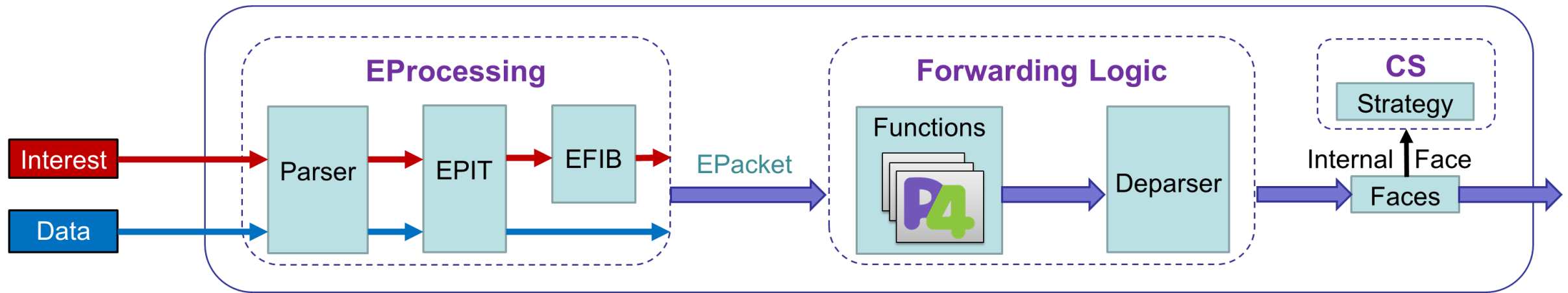
- Design an NDN data plane that allows network operators to associate P4 functions to various NDN namespaces.
- Design a P4 target architecture that allows multiple P4 functions to run in isolation and maintain separate states.
- Allow every P4 function to be reconfigurable separately without any impact on the switch operation.
- Extend the P4 language to allow for efficient string processing while maintaining compatibility with standard P4 compilers.



Proposed Data Plane Architecture



- **Main goal:** perform complex stateful forwarding behaviors on a namespace. Examples: telemetry, geo-fencing, and stateful firewall.





Thank you